

APM1110 FA5 Cellona, Bryan Dion

Github link: https://github.com/bryandioncellona/APM1110/blob/main/FA5/APM1110_FA5_Cellona_BryanDion.Rmd

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1. Discrete Random Variables

A discrete random variable X has the probability mass function

$$P(X = x) = \begin{cases} k(2x + 1), & x = 1, 2, 3, 4, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant k .

Since probabilities must sum to 1,

$$\begin{aligned} \sum_{x=1}^4 k(2x + 1) &= 1, \\ k(3 + 5 + 7 + 9) &= 1, \\ 24k &= 1, \\ k &= \frac{1}{24} \approx \boxed{0.0417}. \end{aligned}$$

```
x <- 1:4
k <- 1 / sum(2 * x + 1)
px <- k * (2 * x + 1)
cat("k =", fmt(k))
```

```
## k = 0.0417
```

(b) Compute the expected value $E[X]$ and the variance $\text{Var}(X)$.

First compute the expected value:

$$\begin{aligned} E[X] &= \sum_{x=1}^4 xP(X=x) \\ &= \frac{1(3) + 2(5) + 3(7) + 4(9)}{24} \\ &= \frac{70}{24} = \frac{35}{12} \approx \boxed{2.9167}. \end{aligned}$$

Next compute the second moment:

$$\begin{aligned} E[X^2] &= \sum_{x=1}^4 x^2P(X=x) \\ &= \frac{1^2(3) + 2^2(5) + 3^2(7) + 4^2(9)}{24} \\ &= \frac{3 + 20 + 63 + 144}{24} = \frac{115}{12} \approx \boxed{9.5833}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \frac{115}{12} - \left(\frac{35}{12}\right)^2 \\ &= \frac{1380 - 1225}{144} = \frac{155}{144} \approx \boxed{1.0764}. \end{aligned}$$

```
EX <- sum(x * px)
EX2 <- sum(x^2 * px)
var_X <- EX2 - EX^2
cat("E[X] =", fmt(EX))
```

```
## E[X] = 2.9167
```

```
cat("E[X^2] =", fmt(EX2))
```

```
## E[X^2] = 9.5833
```

```
cat("Var(X) =", fmt(var_X))
```

```
## Var(X) = 1.0764
```

(c) Calculate $P(X \geq 3)$.

$$P(X \geq 3) = P(X=3) + P(X=4) = \frac{7}{24} + \frac{9}{24} = \frac{2}{3} \approx \boxed{0.6667}.$$

```
p_X_ge3 <- sum(px[x >= 3])
cat("P(X >= 3) =", fmt(p_X_ge3))
```

```
## P(X >= 3) = 0.6667
```

2. Binomial Distribution

An IT system detects that 8% of incoming external requests are malicious. A random sample contains 20 requests. Let B be the number of malicious requests. Assuming independent requests and a constant malicious-request probability,

$$B \sim \text{Binomial}(n = 20, p = 0.08), \quad 1 - p = 0.92.$$

The probability formula is

$$P(B = b) = \binom{n}{b} p^b (1 - p)^{n-b}, \quad \binom{n}{b} = \frac{n!}{b!(n-b)!}.$$

```
n <- 20
p <- 0.08
```

(a) What is the probability that exactly 2 requests are malicious?

$$\begin{aligned} P(B = 2) &= \binom{20}{2} (0.08)^2 (0.92)^{18} \\ &= \frac{20(19)}{2} (0.0064) (0.92)^{18} \\ &= 190(0.0064) (0.92)^{18} \\ &\approx 0.2710906052 \approx \boxed{0.2711}. \end{aligned}$$

```
p_B_eq2 <- dbinom(2, size = n, prob = p)
cat("P(B = 2) =", fmt(p_B_eq2))
```

```
## P(B = 2) = 0.2711
```

(b) What is the probability that at least 3 requests are malicious?

Use the complement of 0, 1, or 2 malicious requests:

$$P(B \geq 3) = 1 - [P(B = 0) + P(B = 1) + P(B = 2)].$$

The component probabilities are

$$\begin{aligned} P(B = 0) &= (0.92)^{20} \approx 0.1886933292, \\ P(B = 1) &= 20(0.08)(0.92)^{19} \approx 0.3281623116, \\ P(B = 2) &\approx 0.2710906052. \end{aligned}$$

Thus,

$$\begin{aligned} P(B \geq 3) &\approx 1 - (0.1886933292 + 0.3281623116 + 0.2710906052) \\ &\approx 0.2120537540 \approx \boxed{0.2121}. \end{aligned}$$

```
# lower.tail = FALSE returns P(B > 2), which equals P(B >= 3).
p_B_ge3 <- pbinom(2, size = n, prob = p, lower.tail = FALSE)
cat("P(B >= 3) =", fmt(p_B_ge3))
```

```
## P(B >= 3) = 0.2121
```

(c) Determine the mean and standard deviation of the number of malicious requests.

$$\begin{aligned} \mu_B &= np = 20(0.08) = \boxed{1.6000}, \\ \sigma_B^2 &= np(1-p) = 20(0.08)(0.92) = 1.472, \\ \sigma_B &= \sqrt{np(1-p)} = \sqrt{1.472} \approx \boxed{1.2133}. \end{aligned}$$

Both the mean and standard deviation are expressed in malicious requests.

```
mean_B <- n * p
sd_B <- sqrt(n * p * (1 - p))
cat("Mean =", fmt(mean_B), "malicious requests")
```

```
## Mean = 1.6000 malicious requests
```

```
cat("Standard deviation =", fmt(sd_B), "malicious requests")
```

```
## Standard deviation = 1.2133 malicious requests
```

3. Poisson Distribution

A server receives an average of 4.5 web traffic hits per minute, with arrivals following a Poisson process. For an interval of t minutes, the expected number of hits is $\lambda = 4.5t$.

For a Poisson random variable H ,

$$P(H = h) = \frac{e^{-\lambda} \lambda^h}{h!}, \quad h = 0, 1, 2, \dots$$

```
rate <- 4.5
lambda_1 <- rate * 1
```

(a) Find the probability of receiving exactly 6 hits in a given minute.

Here $\lambda = 4.5$:

$$\begin{aligned} P(H = 6) &= \frac{e^{-4.5}(4.5)^6}{6!} \\ &= \frac{e^{-4.5}(8303.765625)}{720} \\ &\approx 0.1281201439 \approx \boxed{0.1281}. \end{aligned}$$

```
p_H_eq6 <- dpois(6, lambda = lambda_1)
cat("P(H = 6) =", fmt(p_H_eq6))
```

```
## P(H = 6) = 0.1281
```

(b) Find the probability of receiving at most 2 hits in a given minute.

“At most 2” includes 0, 1, and 2 hits:

$$\begin{aligned} P(H \leq 2) &= \sum_{h=0}^2 \frac{e^{-4.5}(4.5)^h}{h!} \\ &= e^{-4.5} \left(1 + 4.5 + \frac{4.5^2}{2} \right) \\ &= e^{-4.5}(1 + 4.5 + 10.125) \\ &= e^{-4.5}(15.625) \\ &\approx 0.1735780709 \approx \boxed{0.1736}. \end{aligned}$$

```
p_H_le2 <- ppois(2, lambda = lambda_1)
cat("P(H <= 2) =", fmt(p_H_le2))
```

```
## P(H <= 2) = 0.1736
```

(c) Calculate the probability of receiving more than 10 hits in a 2-minute interval.

Let Y count hits in two minutes. Adjust the parameter for the time interval:

$$\lambda_2 = 4.5(2) = 9, \quad Y \sim \text{Poisson}(9).$$

“More than 10” means 11 or more. Use the complement:

$$\begin{aligned} P(Y > 10) &= 1 - P(Y \leq 10) \\ &= 1 - e^{-9} \sum_{y=0}^{10} \frac{9^y}{y!} \\ &= 1 - e^{-9} \left(1 + 9 + \frac{9^2}{2!} + \cdots + \frac{9^{10}}{10!} \right) \\ &\approx 1 - 0.7059883203 \\ &\approx 0.2940116797 \approx \boxed{0.2940}. \end{aligned}$$

```
lambda_2 <- rate * 2
# The upper tail excludes 10, as required by "more than 10".
p_Y_gt10 <- ppois(10, lambda = lambda_2, lower.tail = FALSE)
cat("Two-minute mean =", fmt(lambda_2), "hits")
```

```
## Two-minute mean = 9.0000 hits
```

```
cat("P(Y <= 10) =", fmt(ppois(10, lambda_2)))
```

```
## P(Y <= 10) = 0.7060
```

```
cat("P(Y > 10) =", fmt(p_Y_gt10))
```

```
## P(Y > 10) = 0.2940
```

4. Continuous Probability and Normal Distribution

The processing time T of a complex database query is normally distributed with mean $\mu = 120$ milliseconds and standard deviation $\sigma = 15$ milliseconds:

$$T \sim N(120, 15^2), \quad Z = \frac{T - \mu}{\sigma}.$$

Write $\Phi(z) = P(Z \leq z)$ for the standard normal cumulative distribution function. Do not round the standardized values before calculating probabilities.

```
mu <- 120
sigma <- 15
```

(a) What proportion of queries take longer than 135 milliseconds to process?

$$z = \frac{135 - 120}{15} = 1.$$

Therefore,

$$\begin{aligned} P(T > 135) &= P(Z > 1) = 1 - \Phi(1) \\ &\approx 1 - 0.8413447461 \\ &\approx 0.1586552539 \approx \boxed{0.1587}. \end{aligned}$$

This corresponds to approximately **15.8655%** of queries.

```
p_T_gt135 <- pnorm(135, mean = mu, sd = sigma, lower.tail = FALSE)
cat("P(T > 135) =", fmt(p_T_gt135))
```

```
## P(T > 135) = 0.1587
```

```
cat("Percentage =", fmt(100 * p_T_gt135), "%")
```

```
## Percentage = 15.8655 %
```

(b) Find the probability that a randomly chosen query takes between 100 and 130 milliseconds

Standardize both endpoints:

$$z_L = \frac{100 - 120}{15} = -\frac{4}{3}, \quad z_U = \frac{130 - 120}{15} = \frac{2}{3}.$$

Then

$$\begin{aligned} P(100 < T < 130) &= P\left(-\frac{4}{3} < Z < \frac{2}{3}\right) \\ &= \Phi\left(\frac{2}{3}\right) - \Phi\left(-\frac{4}{3}\right) \\ &\approx 0.7475074625 - 0.0912112197 \\ &\approx 0.6562962427 \approx \boxed{0.6563}. \end{aligned}$$

Since T is continuous, including the endpoints does not change the probability.

```
z_lower <- (100 - mu) / sigma
z_upper <- (130 - mu) / sigma
p_T_between <- pnorm(130, mu, sigma) - pnorm(100, mu, sigma)
cat("Lower z =", fmt(z_lower))
```

```
## Lower z = -1.3333
```

```
cat("Upper z =", fmt(z_upper))
```

```
## Upper z = 0.6667
```

```
cat("P(100 < T < 130) =", fmt(p_T_between))
```

```
## P(100 < T < 130) = 0.6563
```

(c) Find the 95th percentile of query processing times.

Let $t_{0.95}$ satisfy $P(T \leq t_{0.95}) = 0.95$. The standard normal 95th percentile is

$$z_{0.95} = \Phi^{-1}(0.95) \approx 1.6448536270.$$

Solve for the processing time:

$$\begin{aligned} \frac{t_{0.95} - 120}{15} &= z_{0.95}, \\ t_{0.95} &= \mu + \sigma z_{0.95} \\ &= 120 + 15(1.6448536270) \\ &\approx \boxed{144.6728 \text{ milliseconds}}. \end{aligned}$$

Approximately 95% of queries take no more than 144.6728 milliseconds; approximately 5% take longer.

```
z_95 <- qnorm(0.95)
t_95 <- qnorm(0.95, mean = mu, sd = sigma)
cat("Standard normal 95th percentile =", fmt(z_95))
```

```
## Standard normal 95th percentile = 1.6449
```

```
cat("95th percentile =", fmt(t_95), "milliseconds")
```

```
## 95th percentile = 144.6728 milliseconds
```